Vehicle-mounted meters, pump supplied

Category: Volume
Part: 2-IPO
Section: 3
Publication date: 2018-03-15
Revision number: 3

Purpose

This inspection procedure outline (IPO) defines the minimum tests which must be performed to ensure that basic volumetric measuring devices comply with the legislation.

Application

Vehicle-mounted metering systems which incorporate pumping equipment and are used for the measurement of products which are liquid at ambient temperature and pressure (e.g., petroleum products, chemicals, herbicides, pesticides, lubricants, etc.). These metering systems are typically used to deliver liquid products such as fuel to retail and commercial customers.

Equipment

Appropriate volumetric prover standards, thermometer (0.1 °C).

Note: All prover standards must be verified and designated as a local standard by Measurement Canada. Mobile prover standards require recertification every four years. When selecting the appropriate capacity of proving standard, a minimum test volume should equal or exceed a one-minute run at maximum operating rate. Consult bulletin V-24 for additional information.

Requirements

Use

- Device and major components are approved for trade use: Notice of approval (NOA), Weights and Measures Act, section 8
- Device and components are suitable for the actual use: ..... NOA, R.271, R.272
- Device has been initially inspected and is appropriately marked:..... ..... R.29, bulletins V-08 and V-09
- Flow rates are within approved range: .....NOA, R.290
- Automatic temperature compensation (ATC) seasonal and partial conversion): Bulletin V-19
- Volume correction factors (VCFs) are authorized for the actual liquids: .Bulletin V-18, R.236, SVM.2-4

Vehicle-mounted meters, pump supplied

Category: Volume
Part: 2-IPO
Section: 3
Publication date: 2018-03-15
Revision number: 3

- ATC register measurement of two liquids is prohibited if their VCF differs by more than 0.25%: ..... SVM.2-26.

Visual examination (marking and labelling)

- Device is marked with the required information (manufacturer's name, model and serial number, approval number(s), maximum and minimum flow rates, volume corrected to 15 °C, etc.): ..... NOA, R.21.
- Markings are appropriate and permanent (as applicable): R.18
- Marking plate is permanently affixed to the device: R.18
- Initial inspection marks (steel die or approved label): R.29
- Units of measurement and, if applicable, the ¢ price per litre quantity or $ per litre face plate markings are appropriate (location, size, appropriate decimal and number of places): R.135, R.136, R.137
- Marking of multiple outlet systems (if applicable): R.282
- Marking device usage restriction (if applicable): ..... NOA, R.70

Note: The requirement of SVM.2-15 for marking test thermal wells is no longer applied.

Visual examination (printers, tickets, consoles)

- Mandatory interlocked printer on vehicle-mounted meter: R.294
- Tickets are appropriate to the printing device: R.149
- Information is printed as required: ... R.129, R.295, SVM.1-27, SVM.2-27 and bulletin V-20
- Quantity printed with adequate number of digits, decimals: .....R.126, bulletin V-07

Note: A clear message must be printed on the ticket if a delivery is done when the meter is in the testing or calibration mode.

Vehicle-mounted meters, pump supplied

Category: Volume
Part: 2-IPO
Section: 3
Publication date: 2018-03-15
Revision number: 3

Visual examination (seals)

- Calibration adjustments of the meter are sealed: R.235
- Pulse processing adjustments are sealed: SVM.1-8, SVM.1-9
- ATC adjustments are accessible and sealable: SVM.2-7
- Sealing ATC sensor and well assemblies: SVM.2-10
- Sealing flow control valves (if applicable): R.280

Note: Calibration and settings of metrological functions must not be accessible through external devices without breaking seals or without being logged (event logger) as prescribed. The device must not operate normally if left in calibration mode.

Note: The NOA must be consulted for special sealing provisions.

Visual examination (installation – general)

- Device is installed in accordance with restrictions and conditions listed in the NOA and in accordance with the manufacturer's instructions: ..... NOA, R.68, R.69, R.70
- Device is adequately secured, protected against abnormal environmental factors, connected to an adequate and compatible electrical supply as prescribed by the manufacturer, and attached components do not adversely affect its performance: R.124, R.141, R.142
- Primary register is positioned for customer view (if applicable): R.143, R.144
- Minimum graduation is in compliance with requirements: .. R.68(2), R.126, SVM.1-20
- Increments of registration (displayed and printed): ..... Bulletin V-07
- Means of registration of total price – sufficient number of digits: SVM.1-21
- System designed to allow for testing and inspection: R.284
- Blending metering systems – if one meter is equipped with ATC, all meters must be so equipped: SVM.2-27

Visual examination (settings)

- Multiple products or meters – Selection of unit price and adjustment are interlocked: SVM.1-16
- Register used with more than one meter or liquid prevents the delivery of more than one liquid at a time: SVM.1-18

- Multi products ATC – Selection of coefficient of expansion or density is interlocked: SVM.2-12
- Adjacent linearization factors must not deviate by more than 0.25%: SVM.1-10
- Means of adjustments set both gross and net registrations: SVM.2-28

Note: If the factors are interpolated between entered values, the 0.25% restriction between adjacent factors is not applicable.

Note: If the lowest and the highest linearization factor set values exceed 0.25%, additional tests at mid-flow range may be warranted.

Note: The piping of a multi-product / multiple meter system must be equipped with the necessary automatic solenoid valves or other means interfaced with the electronic register to make the interlocking system operational as prescribed by the requirements.

Visual examination (upstream and downstream piping and hoses)

- Short and unencumbered suction piping: R.273
- Piping and accessories are installed so as to minimize the passage of air or vapour: R.274
- Adequate air prevention and elimination system is in place, and the air eliminator has not been obstructed (if applicable): R.276, R.279
- Filter, strainer or other required equipment is located immediately upstream from the meter: .....R.277, NOA
- Sealed flow control valve (if necessary): R.280
- Check valve or other approved means installed to keep the system primed and prevent back flow (if necessary): R.280, R-28, SVM.1-34
- All product is delivered downstream from the transfer point and is retained upstream of the transfer point. Piping and hose downstream of the meter can be readily inspected: R.282
- Multiple outlet systems comply with installation and marking requirements: R.282
- Quick-acting valve installed near outlet for inspection purposes: R.283
- Automatic means (e.g., solenoid valve) to stop liquid flow if missing pulses are detected: SVM.1-37
- Multi-product systems – Interlocked valves to prevent the delivery of more than one liquid at a time: SVM.1-38

Vehicle-mounted meters, pump supplied

Category: Volume
Part: 2-IPO
Section: 3
Publication date: 2018-03-15
Revision number: 3

- Spring-loaded anti-drain valve installed at the outlet end of delivery hose: R.286
- ATC temperature sensor and test well are installed within one metre of the meter ...(if applicable); no components are installed between the ATC sensor and the meter: SVM.2-23, SVM.2-25
- Test thermal well is installed adjacent to the sensor and so that it will retain thermal conducting fluid during a test (if applicable): SVM.2-24
- ATC requirements for blending metering systems composed of more than one meter – all meters must be equipped with ATC: SVM.2-27

Note: Piping downstream from the meter should be closely examined to determine whether product can be redirected during a transaction.

Note: Piping, valves and other components must be examined for leakage. Particular attention must be paid when examining suction piping. For safety and accuracy, any leakage should be corrected before inspection.

Performance

- Certified volumetric provers and measures are required to be wet and dripped in accordance with their certificate of calibration: STP-22
- Volume correction for the expansion or contraction of the prover shell needs to be calculated into the observed volume reading: STP-23
- In-service limits of error apply to tests performed in the field: Bulletins V-03 and V-23
- Return-to-zero test (mechanical registers): STP-1
- Interlock test: STP-2
- Segment test (electronic registers): STP-3
- Slow flow test: STP-4
- Fast flow test: STP-5
- Intermediate flow test: STP-6
- Repeatability test: STP-7
- Out of product test (formerly split compartment test): STP-8
- Agreement between registrations test: STP-9
- Computed value test: STP-10
- Check valve or anti-drain valve test: STP-11

- Hose expansion test (if necessary): STP-12
- Delivery crossover test (if applicable): STP-15
- Product blend test (if applicable): STP-16
- ATC test (if applicable): STP-18
- ATC temperature sensor failure test (if applicable): STP-19
- Electromagnetic interference and radio frequency interference test (if applicable): STP- 20

Note: Annotate inspection certificate with products used during testing and which may be measured for trade use. Ensure programmed calibration factors are applied to appropriate products or product groups.

Note: For meters using an electronic register the following applies:

Meters used to measure more than one liquid must be tested on the highest and lowest grades in terms of viscosity, for each programmed meter factor. Meters equipped with a register having multiple settings for multiple liquids must be interlocked so that the adjustment selection corresponds to the liquid or meter selected.

See specific requirements in specifications SVM.1-16, SVM.1-18, SVM.1-38 and SVM.2-12.

Note: For meters using a mechanical register the following applies:

Mechanical registers will have only one setting for calibration purposes. If the device is used to measure several grades of a product, the meter must be tested on the most viscous product.

Particular products and metering systems

Coriolis effect mass flow meter – Specific requirements

Identify units of measure (mass or volume) to select correct procedures and equipment.

- General requirements: STP-42
- Gravimetric testing/volume indicating: STP-27
- Gravimetric testing/mass indicating: STP-43

Low-flow cut-off

- During initial configuration, low-flow cut-off (LFCO) should be disabled to check suitability of the initial factory applied zero offset.
- LFCO configuration only need be checked if initial configuration parameters have been changed.

- During subsequent examinations, LFCO must be limited to no more than 10% of minimum meter flow rate.
- System must stop flow when actual flow rate is below LFCO.

Initial configuration and examination

- Damping factor set to NOA or manufacturer's recommendations: STP-42
- Determine minimum measured quantity (MMQ): STP-27 or STP-43
- Check Emin suitability (Emin = LOE% × MMQ × 2): .....STP-27 or STP-43
- Limit of error (LOE) applicable at 5 × MMQ or greater (unique to mass flow meters): Section 4.2, Terms and Conditions for the Approval of Coriolis Liquid Meters
- Determine minimum test quantity: .....STP-27, STP-43 or applicable worksheet

Note: Due to the complex format of the uncertainty worksheet, inspectors will typically not have direct access to it. Therefore, STP-27 tables will be applicable unless the Volumetric Specialist has authorized a different value (likely as determined from the uncertainty worksheet).

Note: STP-27 tables should take precedence over the uncertainty worksheet. In most cases, the numbers will be comparable, but concessions were made in the STP tables in order to facilitate the use of standard test equipment already in use. Specifically, there are reduced volumes required for liquid petroleum gas proving with the existing gravimetric provers.

Pipe or piston prover

These standards may be used to test vehicle-mounted meters. Please refer to Measurement Canada's pipe prover training module for complete pipe prover operating procedures.

Metering systems for herbicides, fertilizers and chemicals

Some of these products may require the use of stainless steel provers or pipe provers. Consult the owner of the equipment for safe handling and disposal of the product and cleaning of test equipment before and after the tests. Ensure hoses and seals are compatible with products. Consult the material safety data sheet for safety measures.

See specific requirements in standard test procedure STP-38.

Metering systems for engine oils and other lubricants

As for blending systems, certain meters can be used to measure several grades of lubricants. In this case, they must be tested for accuracy on the two grades that represent the extremes in terms of viscosities and densities. They must perform within applicable tolerances. Meters equipped with a register having multiple settings for multiple liquids must be interlocked so that the adjustment selection corresponds to the liquid or meter selected.

Vehicle-mounted meters, pump supplied

Category: Volume
Part: 2-IPO
Section: 3
Publication date: 2018-03-15
Revision number: 3

See specific requirements in specifications SVM.1-16, SVM.1-18, SVM.1-38 and SVM.2-12.

See specific requirements in standard test procedure STP-25.

- Lubricant meters tested using the gravimetric procedure: STP-27

Lubricant meters tested using volumetric procedure

Volumetric testing requires fully draining, flushing/cleaning with solvent, wetting and dripping the provers (standards) between each test. Take all the necessary precautions to avoid contamination. The device owner must be made aware of the flushing solvent in use and if not acceptable, must provide an alternate agent. An arrangement must be made with the device owner for the safe disposal of the solvent. Field tests are performed using the product currently measured by the meter.

Aircraft refuellers

See specific requirements in standard test procedure STP-36.

Revisions

Revision 3

The purpose of this revision was to add specific tests applicable to Coriolis effect mass flow meters. The format of the document was also changed. Device types and sub-types were removed from the document title.

Revision 2

Notes under the Performance section were expanded to explain what products should be tested with meters using electronic registers and meters using mechanical registers. Minor editorial corrections were also made.

Revision 1

Device types 52.22 and 52.23 were added to the list of device types identified in the title line.